

Quantum Boltzmann Machines

From Quantum Neural Networks to Quantum Generative Models

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Outline

- 1 Recap: Quantum Neural Networks
- 2 Classical Boltzmann Machines
- 3 Quantum Boltzmann Machines: Definition and Structure
- 4 Training Quantum Boltzmann Machines
- 5 Thermofield Double and Purification
- 6 Variational Quantum Boltzmann Machine
- 7 Connections to QNNs and Many-Body Physics
- 8 Restricted Quantum Boltzmann Machine

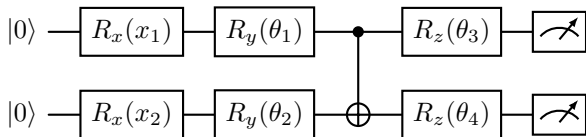
QNN as a Variational Quantum State

The fundamental description of a **Quantum Neural Network**:

$$|\psi(x, \theta)\rangle = U(\theta) U(x) |0\rangle^{\otimes n}, \quad f(x, \theta) = \langle \psi(x, \theta) | O | \psi(x, \theta) \rangle$$

Three building blocks

- $U(x)$: **feature map** — encodes classical data $x \in \mathbb{R}^d$ into a quantum state
- $U(\theta)$: **variational ansatz** — trainable unitary with parameters θ
- O : **observable** — defines the scalar output



Variational Ansatz and Pauli Expansion

The variational unitary is generated by a Pauli Hamiltonian:

$$U(\theta) = e^{-iH(\theta)}, \quad H(\theta) = \sum_{\alpha} \theta_{\alpha} P_{\alpha}, \quad P_{\alpha} \in \{I, X, Y, Z\}$$

The BCH expansion reveals the effective operator:

$$U^{\dagger} O U = O + i[H, O] + \frac{i^2}{2}[H, [H, O]] + \dots$$

Physical meaning: each circuit layer adds one more level of many-body correlations.

Parameter-shift rule (for Pauli generators):

$$\frac{\partial f}{\partial \theta} = \frac{1}{2} [f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})]$$

Quantum Fisher Information

$$F_{ij} = 4 \operatorname{Re}(\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle)$$

Riemannian metric on the variational manifold.

Natural gradient / TDVP

$$\sum_j F_{ij} \dot{\theta}_j = C_i, \quad C_i = \operatorname{Im} \langle \partial_i \psi | H | \psi \rangle$$

Optimal projection of the Schrödinger equation onto the variational manifold.

From Discriminative to Generative Models

Discriminative QNN (what we had)

- Input x encoded via $U(x)$
- Output: class label or regression value
- Minimise: $\mathcal{L} = \sum_k (f(x_k) - y_k)^2$
- Goal: learn $p(y|x)$

Generative QML (our goal now)

- **No** input encoding needed
- Output: full probability distribution $p(x)$
- Minimise: $D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta})$
- Goal: learn to *sample* from p_{data}

Generative models in QML:

- Quantum Boltzmann Machines (QBM)
— *this talk*
- Quantum Circuit Born Machines (QCBM)
- Quantum GANs
- Quantum Variational Autoencoders

Key difference:

QBMs model distributions via a *thermal state* (density matrix) rather than a pure variational state.

Classical Restricted Boltzmann Machine

An RBM has **visible** units $\mathbf{v} \in \{0, 1\}^n$ and **hidden** units $\mathbf{h} \in \{0, 1\}^m$.

Energy function:

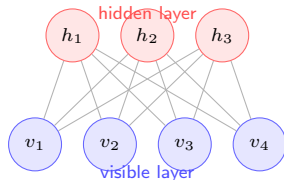
$$E(\mathbf{v}, \mathbf{h}) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} v_i W_{ij} h_j$$

Joint distribution and partition function:

$$p(\mathbf{v}, \mathbf{h}) = \frac{e^{-\beta E}}{Z}, \quad Z = \sum_{\mathbf{v}, \mathbf{h}} e^{-\beta E}$$

Marginal over visible:

$$p(\mathbf{v}) = \frac{1}{Z} \sum_{\mathbf{h}} e^{-\beta E(\mathbf{v}, \mathbf{h})}$$



No intra-layer connections $\Rightarrow$ tractable conditional independence:

$$p(h_j = 1 | \mathbf{v}) = \sigma\left(b_j + \sum_i W_{ij} v_i\right)$$

RBM Training: Contrastive Divergence

Log-likelihood gradient:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

- Data term: easy (one forward pass)
- Model term: hard (requires summing over all $\mathbf{v}, \mathbf{h}$)

Contrastive Divergence (CD- k):

- 1 Clamp $\mathbf{v}^{(0)} = \mathbf{v}_{\text{data}}$
- 2 Run k steps of Gibbs sampling
 $\mathbf{v}^{(0)} \rightarrow \mathbf{h}^{(0)} \rightarrow \mathbf{v}^{(1)} \rightarrow \dots \rightarrow \mathbf{v}^{(k)}$
- 3 Approximate model term with $\mathbf{v}^{(k)}$

CD- k update rule

$$\Delta W_{ij} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_k$$

Similarly for biases a_i, b_j .

Key limitation

The partition function Z is intractable for large systems — gradient estimates are biased. Quantum hardware promises an exact, unbiased estimate of $\langle v_i h_j \rangle_{\text{model}}$ via *quantum sampling*.

From Classical to Quantum Gibbs States

Replace binary units with **qubits** and the classical energy function with a **quantum Hamiltonian**:

Classical: $p(\mathbf{v}) = e^{-\beta E(\mathbf{v})} / Z$

Quantum: The thermal (Gibbs) state

$$\rho(\theta) = \frac{e^{-\beta H(\theta)}}{\text{Tr}[e^{-\beta H(\theta)}]}$$

where $H(\theta)$ is a parametrised quantum Hamiltonian.

The partition function is now a *quantum trace*:

$$Z(\theta) = \text{Tr}[e^{-\beta H(\theta)}] = \sum_k e^{-\beta \epsilon_k(\theta)}$$

where $\{\epsilon_k\}$ are eigenvalues of $H(\theta)$.

Degrees of freedom

Classical RBM	QBM
Binary spins $\{0, 1\}^n$	n qubits
Energy $E(\mathbf{v}, \mathbf{h})$	Hamiltonian $H(\theta)$
Gibbs dist. $e^{-\beta E} / Z$	Gibbs state $e^{-\beta H} / Z$
Marginal $p(\mathbf{v})$	Reduced state ρ_v

Key gain

Quantum off-diagonal (transverse) terms in $H(\theta)$ allow the QBM to represent *quantum correlations* and *superpositions* — beyond any classical BM.

The QBM Hamiltonian

The most general parametrised n -qubit QBM Hamiltonian:

$$H(\theta) = - \sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i - \sum_{i < j} \theta_{ij}^{zz} Z_i Z_j - \dots$$

- θ_i^z : **longitudinal fields** (diagonal) — classical bias terms, analogous to a_i
- θ_{ij}^{zz} : **ZZ couplings** (diagonal) — classical two-body interactions, analogous to W_{ij}
- θ_i^x : **transverse fields** (off-diagonal) — **purely quantum** term, no classical analogue

Thermal state in the eigenbasis

$$H(\theta) |\epsilon_k\rangle = \epsilon_k |\epsilon_k\rangle$$
$$\rho(\theta) = \sum_k \frac{e^{-\beta \epsilon_k}}{Z} |\epsilon_k\rangle \langle \epsilon_k|$$

Classical limit

Setting $\theta_i^x = 0$ for all i recovers a classical BM: $[H, Z_i] = 0 \ \forall i$, so ρ is diagonal in the Z -basis and reduces to a classical Boltzmann distribution.

Visible, Hidden, and Auxiliary Qubits

Following the classical RBM structure, the n qubits are partitioned into:

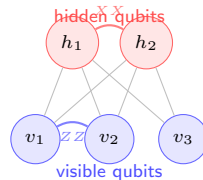
$$\mathcal{H} = \mathcal{H}_v \otimes \mathcal{H}_h, \quad n_v + n_h = n$$

- **Visible** qubits v : coupled to the data; we observe their reduced state $\rho_v = \text{Tr}_h[\rho]$
- **Hidden** qubits h : latent variables that are traced out

The visible marginal is the model distribution:

$$p_\theta(\mathbf{v}) = \langle \mathbf{v} | \rho_v(\theta) | \mathbf{v} \rangle$$

where $\rho_v(\theta) = \text{Tr}_h \left[\frac{e^{-\beta H(\theta)}}{Z} \right]$.



Quantum connections can include XX , YY , ZZ within visible, within hidden, and across layers.

The QBM Training Objective

Given a dataset with empirical distribution $\pi(\mathbf{v})$, we want $p_\theta(\mathbf{v}) \approx \pi(\mathbf{v})$.

Minimise the KL divergence:

$$\mathcal{L}(\theta) = D_{\text{KL}}(\pi \parallel p_\theta) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln \frac{\pi(\mathbf{v})}{p_\theta(\mathbf{v})}$$

Expanding using $p_\theta(\mathbf{v}) = \langle \mathbf{v} | \rho_v | \mathbf{v} \rangle$:

$$\mathcal{L}(\theta) = - \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln p_\theta(\mathbf{v}) + \text{const}$$

$$= - \text{Tr}[\sigma_{\text{data}} \ln \rho_v(\theta)] + \text{const}$$

where $\sigma_{\text{data}} = \sum_{\mathbf{v}} \pi(\mathbf{v}) |\mathbf{v}\rangle \langle \mathbf{v}|$.

Quantum relative entropy

The training objective is the **quantum relative entropy**:

$$\mathcal{L} = S(\sigma_{\text{data}} \parallel \rho_v(\theta))$$

$$= \text{Tr}[\sigma_{\text{data}} (\ln \sigma_{\text{data}} - \ln \rho_v)]$$

This reduces to the classical KL divergence when both states are diagonal.

Full QBM objective

Use the *total* quantum relative entropy:

$$\mathcal{L} = S(\rho_{\text{target}} \parallel \rho(\theta))$$

when the target is itself a quantum state

Gradient of the Log-Likelihood

The gradient of the negative log-likelihood w.r.t. parameter θ_μ is:

$$\frac{\partial \mathcal{L}}{\partial \theta_\mu} = \langle \Lambda_\mu \rangle_{\rho_{\text{clamp}}} - \langle \Lambda_\mu \rangle_{\rho(\theta)}$$

where Λ_μ is the **generalised force** (operator conjugate to θ_μ) defined by:

$$H(\theta) = \sum_{\mu} \theta_{\mu} \Lambda_{\mu}$$

and the two expectation values are:

$$\langle \Lambda_\mu \rangle_{\text{clamp}} = \text{Tr}[\Lambda_\mu \rho_{\text{clamp}}] \quad (\text{data-dependent})$$

$$\langle \Lambda_\mu \rangle_{\theta} = \text{Tr}[\Lambda_\mu \rho(\theta)] \quad (\text{model-dependent})$$

Clamped state

ρ_{clamp} is the Gibbs state of the *same* Hamiltonian but with visible qubits clamped to a data sample:

$$\rho_{\text{clamp}} = \frac{e^{-\beta H^{(v)}}}{Z^{(v)}}$$

where $H^{(v)}$ is $H(\theta)$ with visible-qubit operators replaced by their classical values.

Classical analogy

This is the quantum analogue of:

$$\frac{\partial \ln p}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

The Quantum Log-Likelihood Gradient: Derivation

Key identity. For $H(\theta)$ depending smoothly on θ_μ :

$$\frac{\partial}{\partial \theta_\mu} \ln Z = -\beta \operatorname{Tr}[\Lambda_\mu \rho(\theta)] = -\beta \langle \Lambda_\mu \rangle_\theta$$

Using $p_\theta(\mathbf{v}) = \langle \mathbf{v} | \rho_v | \mathbf{v} \rangle$:

$$\frac{\partial}{\partial \theta_\mu} \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln p_\theta(\mathbf{v}) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \frac{\partial}{\partial \theta_\mu} \ln p_\theta(\mathbf{v})$$

After using the **Golden-Thompson inequality** and the **Duhamel formula** for matrix exponential derivatives:

$$\frac{\partial \mathcal{L}}{\partial \theta_\mu} = \beta [\langle \Lambda_\mu \rangle_{\text{clamp}} - \langle \Lambda_\mu \rangle_\theta]$$

Physical interpretation

Each gradient step reduces the difference between the *clamped* (data-driven) and *free* (model) thermal averages of Λ_μ — the quantum generalisation of Contrastive Divergence.

Challenges: The Partition Function and Thermal Sampling

The intractability problem

Computing $\langle \Lambda_\mu \rangle_\theta = \text{Tr}[\Lambda_\mu \rho(\theta)]$ requires:

$$\rho(\theta) = \frac{e^{-\beta H(\theta)}}{\text{Tr}[e^{-\beta H(\theta)}]}$$

- Diagonalising an $2^n \times 2^n$ matrix — exponential cost classically
- Computing the matrix exponential and its trace
- Repeated for every gradient step

Three main approaches:

- 1 Quantum hardware sampling
- 2 Variational approximation of $\rho(\theta)$
- 3 Semiclassical / mean-field approximations

Quantum hardware advantage

A quantum device *naturally* prepares thermal states via cooling / environment coupling. Measuring $\text{Tr}[\Lambda_\mu \rho]$ requires only $O(\epsilon^{-2})$ shots — no diagonalisation.

Variational approach

Approximate the Gibbs state by a variational ansatz:

$$\rho(\theta) \approx \rho_\phi = U(\phi) \rho_0 U^\dagger(\phi)$$

Jointly optimise (θ, ϕ) — this is the **Quantum-Classical Boltzmann Machine**.

Preparing the Gibbs State on a Quantum Computer

Method 1: Quantum imaginary-time evolution (QITE)

Evolve with the non-unitary operator $e^{-\Delta\tau H}$, approximated by a sequence of unitary gates using TDVP:

$$\rho(\tau) = \frac{e^{-\tau H}}{Z(\tau)}, \quad \tau = \beta/2$$

Each QITE step:

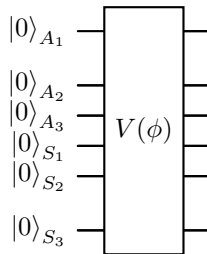
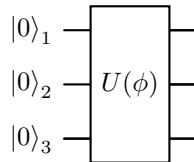
$$e^{-\Delta\tau H} |\psi\rangle \approx U_{\text{QITE}} |\psi\rangle$$

where U_{QITE} is a local unitary found by solving the TDVP linear system $\sum_j F_{ij} \dot{\theta}_j = C_i$.

Method 2: Variational thermal ansatz

$$\rho_\phi = \text{Tr}_A [U(\phi) |0\rangle_{SA} \langle 0| U^\dagger(\phi)]$$

Use ancilla qubits A to purify the mixed state ρ via a



Top: direct variational state. Bottom: thermofield double

Purification and the Thermofield Double State

Any mixed state ρ on a system S can be **purified**: there exists a pure state $|\Psi\rangle_{SA}$ on $S \otimes A$ such that:

$$\rho_S = \text{Tr}_A[|\Psi\rangle\langle\Psi|]$$

For the Gibbs state, the canonical purification is the **thermofield double** (TFD):

$$|\text{TFD}(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_k e^{-\beta\epsilon_k/2} |\epsilon_k\rangle_S |\epsilon_k\rangle_A$$

Tracing out A :

$$\text{Tr}_A[|\text{TFD}\rangle\langle\text{TFD}|] = \rho(\beta)$$

TFD as a QNN circuit

The TFD state can be prepared by a parametrised circuit on $2n$ qubits (n system + n ancilla):

$$|\text{TFD}(\beta)\rangle \approx V(\phi) |0\rangle^{\otimes 2n}$$

with $V(\phi)$ trained to minimise the free energy:

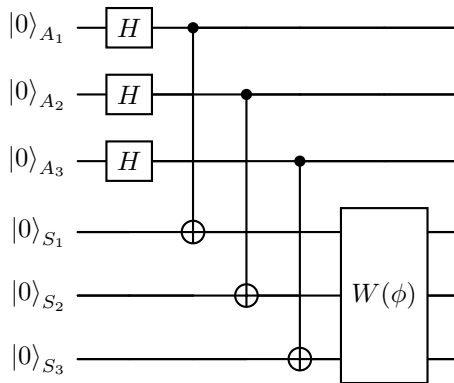
$$\mathcal{F}(\phi) = \langle H \rangle_\phi - \beta^{-1} S_{\text{vN}}(\rho_\phi)$$

where $S_{\text{vN}}(\rho) = -\text{Tr}[\rho \ln \rho]$ is the von Neumann entropy.

Key connection

TFD Circuit: Quantikz Diagram

The thermofield double state prepared on $n = 3$ system qubits + 3 ancilla qubits:



- **Step 1:** $H^{\otimes n}$ on ancillas $\rightarrow$ uniform superposition
- **Step 2:** CNOT from each ancilla to its system partner $\rightarrow$ maximally entangled Bell pairs
- **Step 3:** Variational circuit $W(\phi)$ on system qubits tilts toward the correct thermal distribution

Variational QBM: Full Algorithm

The **Variational QBM** jointly optimises:

- 1 **Model parameters** θ : couplings in $H(\theta)$
- 2 **Thermal circuit parameters** ϕ : prepares $\rho(\beta, \theta) \approx \rho_\phi$

Algorithm (one step):

- 1 Sample $\mathbf{v}$ from dataset
- 2 Run TFD circuit $V(\phi)$ to get $\rho_\phi \approx \rho(\theta, \beta)$
- 3 Clamp visible qubits to $\mathbf{v}$; run $V_{\text{clamp}}(\phi)$ to get ρ_{clamp}
- 4 Estimate $\langle \Lambda_\mu \rangle_\phi$ and $\langle \Lambda_\mu \rangle_{\text{clamp}}$ by sampling
- 5 Update θ by gradient descent
- 6 Update ϕ by imaginary-time TDVP

Two nested optimisations

$$\min_{\theta} \mathcal{L}(\theta) = D_{\text{KL}}(\pi \parallel p_\theta)$$

$$\min_{\phi} \mathcal{F}(\phi, \theta) = \langle H(\theta) \rangle_\phi - \beta^{-1} S_{\text{vN}}(\rho_\phi)$$

Inner: ϕ tracks thermal state of $H(\theta)$.

Outer: θ reduces KL divergence.

Free-energy minimisation

$\mathcal{F}$ is minimised when $\rho_\phi = \rho(\beta, \theta)$ exactly — the quantum analogue of the EM algorithm.

Free Energy and the QBM Objective

The **quantum free energy** at inverse temperature β :

$$\mathcal{F}(\rho, H) = \text{Tr}[H\rho] - \beta^{-1} S_{\text{vN}}(\rho)$$

$$S_{\text{vN}}(\rho) = -\text{Tr}[\rho \ln \rho]$$

The Gibbs state $\rho(\beta) = e^{-\beta H}/Z$ is the *unique* minimiser of $\mathcal{F}$:

$$\rho(\beta) = \arg \min_{\rho} \mathcal{F}(\rho, H)$$

Proof. Using the quantum relative entropy:

$$\mathcal{F}(\rho, H) = \mathcal{F}(\rho(\beta), H) + \beta^{-1} S(\rho \| \rho(\beta)) \geq \mathcal{F}(\rho(\beta), H)$$

since $S(\rho \| \sigma) \geq 0$ (Klein's inequality).

Variational free energy gradient

For a parametrised state ρ_{ϕ} :

$$\frac{\partial \mathcal{F}}{\partial \phi_k} = \text{Tr} \left[H \frac{\partial \rho_{\phi}}{\partial \phi_k} \right] + \beta^{-1} \text{Tr} \left[\ln \rho_{\phi} \frac{\partial \rho_{\phi}}{\partial \phi_k} \right]$$

The second term involves $\ln \rho_{\phi}$, which is expensive classically but accessible via Hadamard-test circuits on quantum hardware.

TDVP for imaginary time

The optimal update of ϕ at each step:

$$\sum_j F_{ij}^{(\phi)} \dot{\phi}_j = -\frac{\partial \mathcal{F}}{\partial \phi_i}$$

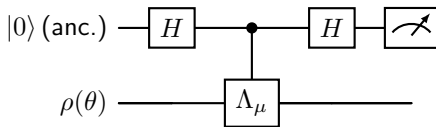
Gradient Computation: Quantum Circuit Approach

Evaluating $\langle \Lambda_\mu \rangle_\theta = \text{Tr}[\Lambda_\mu \rho(\theta)]$:

1. Direct measurement (hardware):

- 1 Prepare $\rho(\theta)$ on the device
- 2 Measure the Pauli observable Λ_μ
- 3 Repeat N_{shots} times; estimate error $\sim 1/\sqrt{N_{\text{shots}}}$

2. Hadamard test (for off-diagonal Λ_μ):



$$\langle \Lambda_\mu \rangle_\theta = 2P(\text{ancilla} = 0) - 1$$

3. Parameter-shift for thermal states

For $\Lambda_\mu = P_\mu$ a Pauli operator:

$$\frac{\partial \langle O \rangle_\theta}{\partial \theta_\mu} = \frac{\beta}{2} [\langle O \rangle_{\theta_\mu + \delta} - \langle O \rangle_{\theta_\mu - \delta}]$$

with $\delta = \pi/(4r)$ where r is the eigenvalue of P_μ : for Paulis $r = 1/2$, so $\delta = \pi/2$.

Important caveat

This thermal parameter-shift rule requires that $\rho(\theta)$ can be re-prepared efficiently for shifted θ — true if the Gibbs state is prepared by a short circuit.

QBM as a Special QNN: Unified View

Both QNNs and QBMs manipulate parameterised quantum states, but the *state type* differs:

Feature	QNN	QBM
State	Pure $ \psi(\theta)\rangle$	Mixed $\rho(\theta)$
Objective	Supervised loss	KL divergence
Gradient	Param-shift	Thermal averages
Key tool	QFI / TDVP	Free energy / TDVP
Sampling	Projective	Thermal / Gibbs
Circuits	Unitary	Unitary + trace

Unifying framework

Both are instances of:

$$\min_{\theta} D(\sigma_{\text{target}} \parallel \mathcal{E}_{\theta}(\rho_0))$$

where $\mathcal{E}_{\theta}$ is a quantum channel (unitary for QNNs, Gibbs map for QBMs) and D is a divergence.

Advantage of QBMs

Mixed-state model $\Rightarrow$ can represent *classically correlated* distributions efficiently, without needing exponentially deep circuits to replicate thermal noise.

Connection to Many-Body Physics

The QBM Hamiltonian lives in the same world as condensed-matter and nuclear physics:

$$H(\theta) = - \sum_{i < j} \theta_{ij}^{zz} Z_i Z_j - \sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i$$

This is the transverse-field Ising model!

- θ_{ij}^{zz} : spin–spin coupling (Ising)
- θ_i^x : transverse field (quantum fluctuations)
- θ_i^z : longitudinal field (external bias)

Thermal state $\rho(\beta) = e^{-\beta H} / Z$ is the *exact* quantum statistical mechanics object studied in condensed matter — QBM training is parameter estimation for a quantum spin model.

Correspondence table

QBM	Many-body physics
Visible qubits	Measured spins
Hidden qubits	Auxiliary fields
θ_i^x	Transverse field
θ_{ij}^{zz}	Ising coupling
$\rho(\beta)$	Thermal density matrix
Training	Inverse problem

Inverse problem

Given samples from $\rho(\beta, \theta^*)$, recover θ^* — equivalent to learning the couplings of a quantum spin Hamiltonian from measurement data.

QBM and the Variational Principle

The QBM free-energy objective

$$\mathcal{F}(\rho_\phi, H(\theta)) = \text{Tr}[H(\theta)\rho_\phi] - \beta^{-1} S_{\text{vN}}(\rho_\phi) \geq \mathcal{F}_{\text{exact}}$$

provides an **upper bound on the free energy** of $H(\theta)$.

Connection to VQE:

VQE minimises $\langle H \rangle_\psi$ over pure states (zero-temperature limit $\beta \rightarrow \infty$). QBM minimises $\mathcal{F}(\rho, H)$ over mixed states (finite temperature).

$$\underbrace{\min_{\psi} \langle \psi | H | \psi \rangle}_{\text{VQE / ground state}} \subset \underbrace{\min_{\rho} \mathcal{F}(\rho, H)}_{\text{QBM / thermal state}}$$

TDVP connection

The thermal gradient for ϕ :

$$\sum_j F_{ij}^{(\phi)} \dot{\phi}_j = -\partial_{\phi_i} \mathcal{F}$$

is the *imaginary-time TDVP* — the same equation used in Stochastic Reconfiguration (SR).

QBM training $\equiv$ SR at finite temperature.

$\beta \rightarrow \infty$ limit

As $\beta \rightarrow \infty$:

$$\rho(\beta) \rightarrow |E_0\rangle \langle E_0|$$

Quantum Natural Gradient for QBM

For the thermal circuit parameters ϕ , the optimal update direction is the **quantum natural gradient**:

$$\Delta\phi = -\eta F^{-1}(\phi) \nabla_{\phi} \mathcal{F}$$

where the QFI metric for *mixed states* is:

$$F_{ij}^{(\phi)} = \text{Tr}[\rho_{\phi} \mathcal{L}_i \mathcal{L}_j]$$

with $\mathcal{L}_i$ the **symmetric logarithmic derivative**:

$$\frac{\partial \rho_{\phi}}{\partial \phi_i} = \frac{1}{2}(\rho_{\phi} \mathcal{L}_i + \mathcal{L}_i \rho_{\phi})$$

This is the **mixed-state QFI** (Bures metric), which generalises the pure-state QFI from week15.

Bures metric vs. Fubini-Study

State type	Metric
Pure $ \psi\rangle$	Fubini-Study $F_{ij}^{\text{FS}} = 4 \text{Re}(\dots)$
Mixed ρ	Bures $F_{ij}^{\text{B}} = \text{Tr}[\rho \mathcal{L}_i \mathcal{L}_j]$

The Bures metric reduces to the Fubini-Study metric for pure states: $F_{ij}^{\text{B}} = F_{ij}^{\text{FS}}$ when $\rho = |\psi\rangle \langle \psi|$.

Practical computation

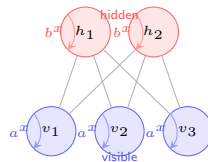
F_{ij}^{B} requires the SLD $\mathcal{L}_i$, which is expensive classically but can be estimated via Hadamard test circuits.

Restricted QBM: Architecture and Hamiltonian

The **Restricted QBM (RQBM)** is the direct quantum analogue of the classical RBM, with n_v visible and n_h hidden qubits:

$$H_{\text{RQBM}} = - \sum_i a_i^z Z_i^v - \sum_j b_j^z Z_j^h - \sum_{ij} W_{ij} Z_i^v Z_j^h - \sum_i a_i^x X_i^v - \sum_j b_j^x X_j^h$$

- a_i^z, b_j^z : classical longitudinal biases
- W_{ij} : inter-layer ZZ coupling (classical part)
- a_i^x, b_j^x : **transverse fields** (quantum)
- No intra-layer ZZ couplings $\Rightarrow$ conditional independence preserved



Classical limit

Set $a_i^x = b_j^x = 0$: recovers exact classical RBM.

Self-loops denote transverse fields a_i^x, b_j^x (quantum fluctuations within each unit).

RQBM Gradient and Contrastive Divergence

For the RQBM, the gradient of the log-likelihood w.r.t. the inter-layer coupling W_{ij} is:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \beta [\langle Z_i^v Z_j^h \rangle_{\text{clamp}} - \langle Z_i^v Z_j^h \rangle_{\text{free}}]$$

For the transverse-field bias a_i^x :

$$\frac{\partial \ln p(\mathbf{v})}{\partial a_i^x} = \beta [\langle X_i^v \rangle_{\text{clamp}} - \langle X_i^v \rangle_{\text{free}}]$$

Key difference from classical: the quantum gradient involves $\langle X_i \rangle$ — an off-diagonal expectation value with no classical analogue.

Quantum CD algorithm

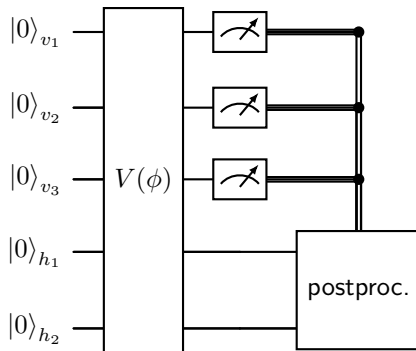
- 1 **Positive phase:** clamp v to data; sample h from $\rho_h^{(v)} = \text{Tr}_v[\rho_{\text{clamp}}]$
- 2 **Negative phase:** run k steps of quantum Gibbs sampling (Metropolis or QITE) to approximate ρ_{free}
- 3 **Update:** $\Delta W_{ij} \propto \langle Z_i Z_j \rangle_+ - \langle Z_i Z_j \rangle_-$

Advantage over classical CD

Quantum sampling of $\langle X_i \rangle$ and $\langle Z_i Z_j \rangle$ is naturally unbiased on quantum hardware — no MCMC convergence issue.

RQBM Circuit: Sampling and Inference

Sampling from the RQBM: prepare the Gibbs state and measure the visible qubits.



Training (positive phase): clamp visible registers to data sample $\mathbf{v}$ and measure $\langle Z_i^v Z_j^h \rangle$, $\langle X_i^v \rangle$, $\langle X_j^h \rangle$.

Training (negative phase): run $V(\phi)$ freely; measure the same observables to estimate the model expectation values $\langle Z_i Z_j \rangle_{\text{free}}$.

Expressivity of QBMs vs. Classical BMs

What a classical RBM can represent

- Any probability distribution over $\{0, 1\}^n$ with sufficiently many hidden units
- But: requires exponentially many hidden units to replicate some quantum distributions
- Limited to distributions arising from diagonal (classical) Hamiltonians

Formal expressivity result:

Any n -qubit density matrix ρ can be expressed as the Gibbs state of some n -qubit Hamiltonian:

$$\rho = \frac{e^{-\beta H}}{Z}, \quad H = -\beta^{-1} \ln(Z\rho)$$

Key consequence: the QBM is a *universal* model for quantum states.

What a QBM adds

- Off-diagonal density matrix elements $\Rightarrow$ coherences
- Entangled marginals over visible qubits
- *Quantum-correlated* probability distributions — not realisable by any classical latent variable

Sample complexity

Learning ρ to within ε in trace distance requires:

$$N = O\left(\frac{n}{\varepsilon^2}\right) \text{ samples}$$

(quantum state tomography lower bound)

When Do QBMs Outperform Classical BMs?

Theoretical separation:

- 1 **Quantum data** (molecular wavefunctions, lattice gauge configs): target is inherently quantum; a QBM learns it directly.
- 2 **Long-range entanglement:** exponentially many classical hidden units needed; $O(n)$ quantum hidden qubits suffice.
- 3 **Partition function estimation:** classical Monte Carlo suffers sign problems for fermionic systems; QBM avoids this.

Practical regime (NISQ):

- Small n ($\lesssim 30$ qubits)
- Shallow thermal circuits (short imaginary time)
- Heuristic advantage — not yet provable

Caution: no-go results

For *classical* data distributions:

- Any QBM can be dequantised (matched by a classical BM) if the quantum correlations of ρ are weak.
- Advantage requires ρ to have genuinely quantum off-diagonal structure.

Barren plateaus in QBMs

Like QNNs, deep QBM circuits suffer exponentially vanishing gradients:

$$\text{Var}(\nabla_{\theta} \mathcal{L}) \sim 2^{-n}$$

Mitigation: shallow circuits and

NumPy Simulation of a Small QBM

```
import numpy as np
from scipy.linalg import expm

def gibbs_state(H, beta=1.0):
    """Exact Gibbs state rho = exp(-beta*H) / Tr[exp(-beta*H)]."""
    evals, evcs = np.linalg.eigh(H)
    weights = np.exp(-beta * evals)
    weights /= weights.sum()
    rho = sum(w * np.outer(v, v.conj())
              for w, v in zip(weights, evcs.T))
    return rho

def qbm_hamiltonian(theta_z, theta_x, W, n_v, n_h):
    """Transverse-field Ising RBM Hamiltonian."""
    n = n_v + n_h
    I2 = np.eye(2); Z = np.diag([1., -1.]); X = np.array([[0., 1.], [1., 0.]])
    def kron_op(gate, i):
        ops = [gate if j == i else I2 for j in range(n)]
        M = ops[0]
        for op in ops[1:]: M = np.kron(M, op)
        return M
    H = np.zeros((2**n, 2**n), dtype=complex)
    for i in range(n):
        H -= theta_z[i] * kron_op(Z, i)
    for i in range(n):
        H -= theta_x[i] * kron_op(X, i)
    for i in range(n_v):
        for j in range(n_h):
```

QBM Training Loop (NumPy)

```
def qbm_gradient(theta_z, theta_x, W, data, n_v, n_h, beta=1.0):
    """Estimate gradient of negative log-likelihood."""
    n = n_v + n_h
    I2 = np.eye(2); Z = np.diag([1., -1.])
    H = qbm_hamiltonian(theta_z, theta_x, W, n_v, n_h)
    rho_free = gibbs_state(H, beta)

    dW = np.zeros_like(W)
    for v_sample in data:
        # Clamp visible qubits to data sample
        H_clamp = clamp_hamiltonian(H, v_sample, n_v)
        rho_clamp = gibbs_state(H_clamp, beta)
        for i in range(n_v):
            for j in range(n_h):
                ZZ = kron_op(Z, i, n) @ kron_op(Z, n_v+j, n)
                pos = np.real(np.trace(ZZ @ rho_clamp))
                neg = np.real(np.trace(ZZ @ rho_free))
                dW[i, j] += beta * (pos - neg)
    return dW / len(data)

# Training loop
lr = 0.05
for step in range(200):
    grad_W = qbm_gradient(theta_z, theta_x, W, data, n_v, n_h)
    W += lr * grad_W
    if step % 50 == 0:
```

- ① **Classical limit.** Set $\theta_i^x = 0$ for all i in the RQBM. Show that the Gibbs state becomes diagonal in the Z -basis and the gradient reduces to the classical CD update.
- ② **Free energy minimisation.** For a single qubit with $H = -\theta^x X - \theta^z Z$, compute the exact Gibbs state $\rho(\beta)$, the free energy $\mathcal{F}(\rho, H)$, and verify that $\rho(\beta)$ minimises $\mathcal{F}$ over all single-qubit density matrices.
- ③ **TFD circuit.** Implement the TFD preparation circuit for $n = 2$ qubits using NumPy. Verify that tracing out the ancilla reproduces the Gibbs state up to circuit approximation error.
- ④ **Bures vs. Fubini-Study.** For a pure state $\rho = |\psi\rangle\langle\psi|$ show analytically that $F_{ij}^B = \text{Tr}[\rho \mathcal{L}_i \mathcal{L}_j]$ reduces to $F_{ij}^{\text{FS}} = 4 \text{Re}(\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle)$.
- ⑤ **Quantum advantage.** Construct a 2-qubit density matrix ρ with off-diagonal coherences not representable by any classical BM. What is the minimum number of hidden qubits needed?

Outlook: Open Problems and Future Directions

Near-term (NISQ) focus

- Small QBMs ($n \lesssim 30$) on current hardware
- Hybrid: quantum sampling + classical update
- Applications: quantum chemistry, finance, combinatorial optimisation
- Error mitigation for thermal state preparation

Key open questions

- Can QBMs learn classically hard-to-sample distributions efficiently?
- Can barren plateaus be avoided with problem-inspired structure?
- What is the quantum sample complexity advantage over classical BMs?

Long-term directions

- **Fault-tolerant QBMs:** exact Gibbs state prep via quantum signal processing
- **Quantum GANs:** replace discriminator or generator with a QBM
- **Physics:** learn Hamiltonians from tomographic data; phase diagram estimation; sign-problem-free MC via QBM proposals
- **Tensor networks:** finite- T DMRG $\leftrightarrow$ shallow QBM circuits

Quantum advantage landscape:



Summary: QBM in Context

	Classical BM	QNN	QBM
State	Distribution $p(\mathbf{v})$	Pure $ \psi(\theta)\rangle$	Mixed $\rho(\theta)$
Model	$e^{-\beta E}/Z$	$U(\theta) 0\rangle$	$e^{-\beta H(\theta)}/Z$
Gradient	CD (biased)	Param-shift (exact)	Thermal avg. (exact on QC)
Geometry	Euclidean	Fubini-Study QFI	Bures metric
Training	Gibbs sampling	TDVP (real time)	TDVP (imag. time)
Expressivity	Classical dists.	Pure states	All quantum states
Advantage	Tractable	Discriminative	Generative

The central message

A Quantum Boltzmann Machine is a **thermal QNN**: it replaces the pure-state variational ansatz with a Gibbs (mixed) state, enabling universal generative modelling of quantum distributions with a natural connection to statistical mechanics and the variational principle.

Quantum Boltzmann Machines

- ❶ M. H. Amin et al., *Quantum Boltzmann machine*, Phys. Rev. X **8**, 021050 (2018).
- ❷ G. Verdon et al., *A quantum algorithm to train neural networks using low-depth circuits*, arXiv:1712.05304 (2017).
- ❸ J. Gao & L.-M. Duan, *Efficient representation of quantum many-body states with deep neural networks*, Nat. Commun. **8**, 662 (2017).
- ❹ C. Zoufal et al., *Variational quantum Boltzmann machines*, Quantum Mach. Intell. **3**, 7 (2021).

Thermofield double and Gibbs state preparation

- ❺ S. Chowdhury et al., *A variational quantum algorithm for preparing quantum Gibbs states*, arXiv:2002.00055 (2020).
- ❻ D. Zhu et al., *Generation of thermofield double states and critical ground states with a quantum computer*, PNAS **117**, 25402 (2020).

QNN and QFI (see also week15)

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- ❽ M. H. Amin, *Searching for quantum speedup in quasistatic quantum annealers*, Phys. Rev. A **92**, 052323 (2015).
- ❾ R. Babbush et al., *Focus beyond quadratic speedups for error-corrected quantum advantage*, PRX Quantum **2**, 010103 (2021).